Multimedia Signal Processing - 2nd Module

July, 12 2016

1. **Question (8 pts)**

   Consider a continuous source X that emits symbols following a probability distribution function (p.d.f.) $p_X(x)$.

   (a) Explain how to design the Lloyd-Max quantizer. Consider $M = 3$ quantization bins and use the thresholds $\{-\infty, a_2, a_3, +\infty\}$ as initialization.

   (b) Consider the following series of symbols emitted by X

   $$x = \{-0.9,\ 0.1,\ -0.8,\ -0.7,\ 0.7,\ -0.6,\ -0.2,\ 0.1,\ 0.7,\ -0.2,\ 1.0\}.$$

   Design a Lloyd-Max quantizer tailored to these symbols using $M = 3$ bins and the thresholds $\{-\infty, -0.5, 0.5, +\infty\}$ as initialization. Stop after one complete iteration.

   (c) How many iterations must be performed to reach the algorithm's convergence?

   (d) Quantize x using the Lloyd-Max quantizer designed at point (b), thus obtaining $\tilde{x}$.

   (e) Compute the entropy of the source representing $\tilde{x}$ estimating the probability mass function (p.m.f.) directly from $\tilde{x}$.

   (f) Derive an Huffman code to transmit $\tilde{x}$. Represent using this code a string of four symbols of your choice.

2. **Question (5 pts)**

   (a) Illustrate the block diagram of a hybrid DCT/DPCM video encoder.

   (b) Illustrate the block diagram of a hybrid DCT/DPCM video decoder.

3. **Question (5 pts)**

   (a) Illustrate the principles of transform coding.

   (b) Provide its geometric interpretation.

4. **Questions (5 pts - each answer can be either TRUE or FALSE)**
   (a) Consider an image coding scheme.
   - [T] [F] Each color channel in the RGB space is completely independent on natural images.
   - [T] [F] It is better to sub-sample chroma components than luminance component.
   - [T] [F] Neighboring pixels typically share some form of correlation.
   - [T] [F] KLT is used for JPEG compression.

   (b) Consider an audio coding scheme.
   - [T] [F] The concept of critical bandwidth is only used for speech coding.
   - [T] [F] MPEG-1 psychoacoustic model exploits both tonal and noise maskers.
   - [T] [F] The absolute threshold of hearing has a minimum around 3 or 4 kHz.

   (c) Consider two independent discrete memoryless sources X and Y.
   - [T] [F] $H(X|Y) + H(Y) = H(X,Y)$.
   - [T] [F] The average length of the Huffman code L_X is always greater than $H(X)$.
   - [T] [F] The maximum distortion achievable quantizing X is given by the source variance σ_X^2.

5. **Question (Matlab - 10pts)**
   (a) Load the file `table_tennis.mat`, which contains two frames of a video sequence (i.e., `table_tennis(:,:,1)` and `table_tennis(:,:,2)`). Display both frames on the screen and describe what they represent.

   (b) Consider each frame as a monodimensional realization generated by a discrete memoryless source. Compute the entropy of each frame separately.

   (c) Consider splitting each frame into 8×8 non-overlapping blocks. For each block of the first frame, compute a motion vector (`MV = [x y]`) by minimizing the mean squared error (MSE) with a block of the second frame. Only consider a neighborhood of ± 1 blocks with respect to the reference position (i.e., at most 9 comparisons per reference block).

   (d) Compute the entropy of the MVs considering: i) the horizontal component only (i.e., `x`); ii) the vertical component only (i.e., `y`); iii) both components jointly (i.e., `[x y]`).

   (e) Given the MVs, compute and display the displaced frame difference (DFD).

   (f) As in point (b) compute the entropy of the DFD.

   (g) Considering that you work with an optimum code, which is the minimum number of bits needed to transmit both frames: i) when each one is transmitted separately; ii) when you transmit only one frame, the DFD and the MVs.